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Unit – 2 : Session – 6 : SLO – 2 | Binary Arithmetic – Multiplication and Division

■ Section A – Match the Following (Answers)

1. Binary Multiplication → B. Use 0/1 multiplication and shifting
2. Binary Division → A. Repeated subtraction and shifting
3. Multiplying by 0 → C. Always gives 0
4. Multiplying by 1 → D. Gives the same number
5. Shift Left → E. Equivalent to multiplying by 2

■ Section B – Fill in the Blanks

- Binary multiplication uses only two operations: AND and SHIFT LEFT.
- Division in binary uses repeated SUBTRACTION with right shifts.
- Shifting left by one bit is equal to MULTIPLYING by 2.
- Shifting right by one bit is equal to DIVIDING by 2.
- If divisor is greater than the dividend, the quotient is 0 and the remainder is the DIVIDEND.

■ Section C – Activity Task (Binary Multiplication)

- (a) 1011×0101 : partial products $1011 (\times 1)$, $0000 (\times 0)$, 1011 shifted 2 ($\times 1$), 0000 shifted 3 ($\times 0$); sum = 0110111 . Answer: $1011 \times 0101 = 0110111 (= 11 \times 5 = 55)$.
- (b) 1101×0011 : partial products $1101 (\times 1)$, 1101 shifted 1 ($\times 1$), 0000 , 0000 ; sum = 100111 . Note: $1101 \times 0011 = 13 \times 3 = 39 = 0100111$; adding shifted partial products $1101 + 11010 = 100111 \rightarrow$ Answer: $1101 \times 0011 = 100111$.

■ Section D – Activity Task (Binary Division)

- (a) $10110 \div 101$: 101 goes into 1011 once (quotient bit 1), remainder 0110 brings down; $101 > 011 \rightarrow$ quotient bit 0; final: Quotient = 100, Remainder = 10. Check: $10110 \blacksquare = 22$, $101 \blacksquare = 5 \rightarrow 22 = 5 \times 4 + 2$, so Quotient = 100 (4), Remainder = 010 (2).
- (b) $11001 \div 11$: 11 into 110 twice (11), $11001 = 25$, $11 \blacksquare = 3 \rightarrow 25 = 3 \times 8 + 1$, so Quotient = 1000 (8), Remainder = 1. Long division: 11 into 110 $\rightarrow 1$ (rem 0), bring 0 $\rightarrow 00 < 11 \rightarrow 0$, bring 1 $\rightarrow 001 < 11 \rightarrow 0$, final remainder 1 $\rightarrow$ Quotient = 1000, Remainder = 001.

■ Section E – True or False

- Binary multiplication uses only 0, 1 and shifting. — TRUE
- Division by shifting right is exact for any binary number. — FALSE (exact only when no 1-bits are shifted out, e.g. even numbers)
- In binary multiplication, multiplying by 1 leaves the number unchanged. — TRUE
- In binary division, if the divisor is bigger than the partial dividend, a 0 is placed in the quotient. — TRUE
- Shifting left by two positions multiplies the number by 4. — TRUE